Claude Code for Accounting Research

Module 6 — Seminar: AI, the Production of Knowledge, and the Academic Research Market

John Barrios

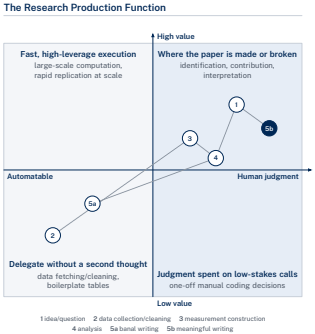
Yale School of Management

The Week, With a Price Tag

Module	What you built	This week	∼2019
2	R&D-intensity panel, SEC FSDS	80 min lab	RA, ∼1 week
3	EDGAR risk-factor pipeline + reusable skill	80 min lab	RA, 2–3 weeks
4	Filtered, verified Compustat `funda` pull	80 min lab	RA, several days
5	Hardened, rerunnable repo: paper trail, git	∼3 hrs	RA, days of audit

That speed-up raises a question that is not technical.

Place Your Dissertation on This Map

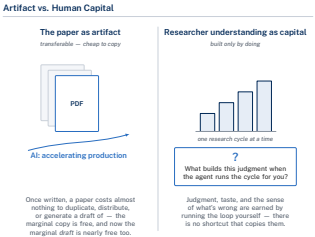


Pipeline stages $\times$ automatable$\leftrightarrow$human-judgment $\times$ low$\leftrightarrow$high value.

Do this now, on your own

Place your own dissertation's stages on this map. Don't discuss yet. Block 1 opens with two volunteers defending their placement.

The Artifact and the Capital



Artifact: transferable, cheap to copy. Capital: built only by doing the work.

The personal audit

Can you discuss it without the LLM? Bastani et al.: GPT-4 raised practice grades 48%; take it away and those students scored 17% worse. A guardrail tutor largely removed that harm. Budzyń et al.: endoscopists' unaided detection rate fell from 28.4% to 22.4% after AI arrived (before/after, no control group).

Provenance: Who Is Accountable?

"A computer can never be held accountable."
— attributed to an IBM training slide, 1979

The journals' two rules (*Nature* 2023; versions now at the AER and AAA)

1. No LLM will be credited as an author. Authorship carries accountability a model cannot take on.
2. Any use of these tools must be disclosed

A Retraction, Not a Citation

Retraction case study: labeled as such, never cited as evidence

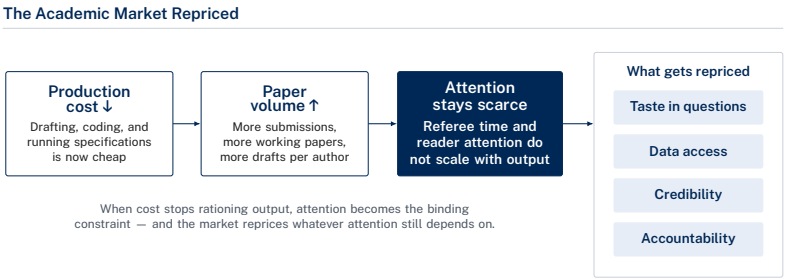
A 2024 working paper claimed that a randomized rollout of an AI materials-discovery tool led researchers to discover 44% more new materials

- Circulated to acclaim from prominent economists, then MIT said it had “no confidence in the provenance, reliability or validity of the data”
- MIT asked for the paper to be withdrawn; the author is no longer at MIT
- The field’s provenance checks failed *before* AI was writing papers
- Accounting’s case: Kim, Muhn & Nikolaev (2024) temporarily withdrawn by the authors in 2025 after a co-author’s replication found inconsistencies

The open question

Does AI make that machinery stronger (audit trails) or weaker (volume)?

Papers Are the Currency



Placement, tenure, prestige, salary all clear through publications. Kusumegi et al.: LLM adopters post 24–89% more papers, and polished prose stops signaling quality. Hao et al.: papers using AI methods (mostly pre-LLM) go with 3× more output per scientist and a narrower range of topics.

Rules of Engagement

- Three blocks, $\sim$30 min each, with a 5-min stretch between
- Each block: 2-min position statements (assigned in advance) $\rightarrow$ open discussion $\rightarrow$ instructor synthesis
- No catastrophizing and no dismissing: pair every provocation with its counterargument
- Memo due one week after today

Block 1 — Knowledge vs. Learning

- If AI produces the artifact, what happens to the learning?
- Opens with your dissertation placement; two volunteers defend theirs
- Which pipeline stages build intuition worth protecting?
- Accounting parallel: AI-assisted audit judgment

Closing question

Is PhD training a production process, or a transformation of the student?

Block 2 — Man vs. Machine

- Does it matter, epistemically or socially, who or what produced a result?
- Two identical extractions, one hand-coded by an RA and one from Claude with an audit trail. Are they equivalent?
- A spotless audit trail can still deliver a wrong estimate
- 10-min segment: when a second model reviews the first model's code, what is verified?

Closing question

Does AI make the field's provenance-checking machinery stronger or weaker?

Block 3 — Returns to Research

- Papers are the currency. What happens when production costs collapse?
- You built the Module 4 pipeline in an 80-minute lab. Suppose every accounting PhD can.
- Scarce candidates: question taste, data access and institutional knowledge, credibility, accountability
- Homogenization: if every referee report passes through the same models, does voice still signal anything?

Closing question

Name one input you believe stays scarce. That answer is the starting point for your memo.

The Memo

"What I will still be paid for in 2036": one page, due in one week

- Commit to specific components of the research production function that stay human
- Defend with at least one assigned reading *and* one experience from the build modules
- Name one concrete investment for your remaining PhD years
- AI help allowed except for the argument; one-line disclosure

Grading

Credit/no-credit on **defended specificity**. An optimist and a pessimist can both earn credit; a vague memo cannot.